📘 Playwright Locator Notes (CSS Selectors)

```
import { test } from "@playwright/test";

test("Verify Locator", async ({ page }) => {

    await page.goto("https://gmail.dev/");

    // Using Tag Name + Attribute
    await page.locator('input[id="identifierId"]').fill("RBG Technologies");

    // Using Attribute only
    await page.locator('[id="identifierId"]').fill("RBG Technologies");

    // Using Class Selector
    await page.locator('.whsOnd.zHQkBf').nth(0).fill("RBG Technologies");

    // Using Id Selector only
    await page.locator(#identifierId).fill("RBG Technologies");

    await page.pause();

});
```

CSS Selector Syntax

1. Tag Name + Attribute

Syntax

```
tagName[attribute='attributeValue']
```

Example

```
input[id='identifierId']
```

Explanation

```
input
   |
   ▼
Tag Name

[id='identifierId']
        |
        ▼
Attribute = Attribute Value
```

Meaning

- Find an **input** element
- Whose **id** is **identifierId**

Example

```
await page.locator("input[id='identifierId']").fill("RBG Technologies");
```

---

2. Attribute Only

Syntax

```
[attribute='attributeValue']
```

Example

```
[id='identifierId']
```

Explanation

```
No Tag Name

Search any element

having

id='identifierId'
```

Example

```
await page.locator("[id='identifierId']").fill("RBG Technologies");
```

When should we use this?

- When the attribute value is unique.
- No need to mention the tag name.

---

3. ID Selector

Syntax

```
#attributeValue
```

Example

```
#identifierId
```

Equivalent to

```
[id='identifierId']
```

Example

```
await page.locator("#identifierId").fill("RBG Technologies");
```

Explanation

```
#

means

ID Selector
```

It is the shortest and fastest way to locate an element by its **id**.

---

4. Class Selector

Syntax

```
.className
```

Example

```
.whsOnd
```

Meaning

Find elements having

```
class="whsOnd"
```

---

5. Multiple Class Selector

Syntax

```
.class1.class2
```

Example

```
.whsOnd.zHQkBf
```

Meaning

Find elements containing **both** classes.

Example HTML

```
<input class="whsOnd zHQkBf">
```

Playwright searches for

```
class contains

whsOnd

AND

zHQkBf
```

---

6. nth()

Example

```
await page.locator(".whsOnd.zHQkBf").nth(0).fill("RBG Technologies");
```

Why use `nth()`?

If multiple elements match the same locator,

Playwright returns all matching elements.

`nth(index)` selects one element based on its position.

```
0 → First Element

1 → Second Element

2 → Third Element
```

Which Locator Should We Prefer?

✅ 1st Preference

```
#identifierId
```

Reason

- Short
- Fast

- Unique (if the id is unique)

✅ 2nd Preference

```
[id='identifierId']
```

Reason

- Useful when you want to locate by attribute.
- Works even without mentioning the tag name.

✅ 3rd Preference

```
input[id='identifierId']
```

Reason

- More specific.
- Useful when multiple elements have similar attributes.

✅ 4th Preference

```
.whsOnd.zHQkBf
```

Reason

- Use when there is no unique id or stable attribute.
- Be careful because class names can change frequently.

Locator Priority (Best Practice)

```
1. ID
   ↓
#identifierId

2. Attribute
   ↓
```

```
[id='identifierId']

3. Tag + Attribute
   ↓
input[id='identifierId']

4. Class
   ↓
.whsOnd.zHQkBf

5. nth()
   ↓
Use only when multiple elements match.
```

Interview Questions

Q1. What is the difference between

```
input[id='identifierId']
```

and

```
[id='identifierId']
```

Answer:

- `input[id='identifierId']` searches only `<input>` elements with that id.
- `[id='identifierId']` searches **any HTML element** with that id.

Q2. What does `#identifierId` mean?

Answer:

It is a CSS **ID Selector**.

Equivalent to:

```
[id='identifierId']
```

Q3. What does `.whsOnd.zHQkBf` mean?

Answer:

It is a **multiple class selector**.

The element must contain **both** classes:

- `whsOnd`
- `zHQkBf`

Q4. Why do we use `nth(0)`?

Answer:

When multiple elements match the same locator, `nth(0)` selects the **first matching element**.